



REVIEW ARTICLE

The molecular pathogenesis of the myelodysplastic syndromes

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Abstract

Recent studies have greatly illuminated the genomic landscape of the myelodysplastic syndromes (MDS), and the pace of discovery is accelerating. The most common mutations found in MDS occur in genes involved in RNA splicing (including *SF3B1*, *SRSF2*, *U2AF1*, and *ZRSR2*) and epigenetic modification (including *TET2*, *ASXL1*, and *DNMT3A*). The identification of spliceosome mutations in approximately half of all patients with MDS implicates abnormalities of RNA splicing, a pathway not previously known as a target for mutation, in the MDS pathogenesis. Several regulators of signal transduction (*NRAS*, *JAK2*) and transcription factors (*RUNX1*, *TP53*) are also frequently mutated in MDS. The complex patterns of associations between gene mutations identified have revealed epistatic interactions between spliceosome components and epigenetic modifiers in MDS. The cytogenetic abnormalities found in MDS are characterized by the loss of genetic material, whereas translocations are rare. The cytogenetic deletion maps of MDS (e.g., 5q-, 7q-, 20q-) provide us with circumstantial evidence for the presence of tumor suppressor genes. It is now recognized that haploinsufficiency (a gene dosage effect) resulting from gene deletions or inactivating mutations is an important disease mechanism in MDS. Haploinsufficiency of the ribosomal protein gene *RPS14* plays a critical role in the development of anemia in the 5q- syndrome, and haploinsufficiency of *CUX1* is important in some patients with MDS and AML with complete or partial loss of chromosome 7. Gene expression profiling has identified key deregulated genes and pathways and new prognostic gene signatures in MDS. Recent advances in the molecular pathogenesis of MDS are leading to new biological, clinical, and therapeutic insights.

Key words Myelodysplastic syndromes; gene mutation; molecular pathogenesis; gene expression**Correspondence** Dr Andrea Pellagatti, LLR Molecular Haematology Unit, Nuffield Division of Clinical Laboratory Sciences, Radcliffe Department of Medicine, University of Oxford, John Radcliffe Hospital, Oxford OX3 9DU, UK. Tel: +44 1865 222911; Fax: +44 1865 221778; e-mail: andrea.pellagatti@ndcls.ox.ac.uk

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The myelodysplastic syndromes (MDS) are a heterogeneous group of clonal hematopoietic stem cell (HSC) malignancies characterized by ineffective hematopoiesis and peripheral blood cytopenias, and patients typically have a hypercellular bone marrow. Patients with MDS show frequent progression (approximately 40% of cases) to acute myeloid leukemia (AML) (1). Over the past decade, the application of new high-throughput technologies, including SNP-array analysis and next-generation sequencing (NGS), to the study of MDS has led to the identification of several recurrently mutated genes in this disorder (2, 3). For example, the identification of submicroscopic deletions or regions of acquired uniparental

disomy has provided important clues that resulted in the discovery of novel gene mutations (4, 5). Moreover, new insights into the genetic basis of MDS have been gained in recent years with the use of whole-genome sequencing and whole-exome sequencing (WES) (6–8). Some of the mutated genes identified using these approaches have been shown to provide important prognostic information. A seminal paper by Bejar *et al.* (9) reported that mutations in *ASXL1*, *TP53*, *EZH2*, *ETV6*, and *RUNX1* are predictors of poor overall survival in patients with MDS, independently of established risk factors, and these findings have been subsequently confirmed in several studies. Microarray-based gene expression profiling has

identified many differentially expressed genes in MDS and highlighted several critical gene pathways deregulated in this disorder (10, 11). The development of new technologies has clearly been a key driver of recent discoveries in MDS. In this review, we discuss the molecular pathogenesis of MDS, with an emphasis on recent advances in the field.

Gene mutations in MDS

Recurrent somatic mutations in several genes have been reported in MDS and implicated in the development and/or progression of this disorder. Table 1 shows a list of commonly mutated genes in MDS.

Splicing factor mutations

The recent discovery of somatic spliceosomal mutations in MDS has revealed a new leukemogenic pathway involving spliceosomal dysfunction. In 2011, Papaemmanuil *et al.* (6) identified frequent mutation of the splicing factor gene *SF3B1* in low-risk MDS using whole-exome sequencing (WES) analysis. *SF3B1* mutations were found in >70% of patients whose disease is characterized by ring sideroblasts (6, 12). The close association between *SF3B1* mutation and the ring sideroblast phenotype is consistent with a causal link; *SF3B1* is the first gene to be strongly associated with a specific feature of MDS. Up to 28% of all patients with MDS harbor *SF3B1* mutations (6, 12, 13), making *SF3B1* the most commonly mutated gene found in MDS to date. In parallel, Yoshida *et al.* (8) also identified recurrent mutations of *SF3B1* and other genes encoding members of the RNA splicing machinery, including *U2AF35* (*U2AF1*), *SRSF2*, and *ZRSR2* in MDS using WES. These spliceosome mutations were frequent in MDS (occurring in 130/228 patients; 57%), and highly specific to this disorder. Mutations in *SRSF2*, *U2AF1*, and *ZRSR2* have been found in approximately 12–15%, 7–9%, and 3–11% of patients with MDS, respectively (8, 13). Most of the spliceosome mutations are mutually exclusive, and the affected genes are involved in the 3' splice site recognition step of pre-mRNA processing (8). Mutations in other splicing factor genes, *PRPF40B*, *SF3A1*, *SF1*, and *U2AF65* (*U2AF2*), are less frequent, each occurring in approximately 1–2% of patients with MDS.

Most splicing factor gene mutations identified in MDS to date are missense and affect rather invariant positions in *SF3B1*, *SRSF2*, *U2AF1*, and other spliceosomal proteins (8). The presence of mutational hot spots and the absence of frameshift or nonsense changes suggest that the large majority of the splicing factor mutations are gain-of-function or change-of-function (neomorphic) mutations (6, 8, 14, 15). A recent study described a *Sf3b1* heterozygous knockout mouse showing the presence of ring sideroblasts in the bone marrow, suggesting that haploinsufficiency of *SF3B1* may be the relevant mode of action of *SF3B1* mutations in MDS

(16). Recent similar studies have not made this observation, however (17, 18). Notably, none of the mice described in these studies developed MDS (16–18). In the case of *ZRSR2*, missense, nonsense, and frameshift mutations have been identified and are considered to be loss-of-function (8, 15). As most spliceosomal mutations are considered to result in a gain-of-function or neomorphic phenotype, treatment with specific spliceosomal inhibitors could allow for targeting of the mutant phenotype (19).

The major targets of spliceosome mutations in MDS are components of the E/A splicing complex, which plays a role in the initial steps of RNA splicing (8). The frequency and specificity of these spliceosome mutations, and the fact that they occur in a mutually exclusive manner, suggest that the abnormal function of the E/A splicing complex is a major feature in MDS (8, 13).

Given the crucial function of the E/A splicing complex on the recognition of 3' splice sites, the likely consequence of the spliceosome mutations observed in MDS is impaired/ altered splicing of different RNA transcripts. The disruption of 3' splice site recognition could result in altered exon usage or activation of cryptic 3' splice sites. Abnormal splicing may ultimately lead to decreased expression resulting in loss of function of key downstream target genes, including tumor suppressor genes (TSGs).

The target mRNAs affected by splicing factor mutations in MDS are beginning to be identified. Recent RNA sequencing (RNA-seq) data from bone marrow cells of a small number of MDS patients with spliceosome mutations suggest that these mutations do have effects on specific genes, including several that have been previously shown to play an important role in MDS pathogenesis. For example, aberrant splicing of *RUNX1* was observed in two patients with *SRSF2* mutations (20) and of *ASXL1*, *EZH1*, and *CBL* in two patients with *SF3B1* mutations (16). Higher expression of a specific isoform of *SLC25A37*, a crucial importer of Fe²⁺ into the mitochondria of erythroid precursors, has recently been described in *SF3B1* mutant patients (21). Moreover, many genes significantly differentially expressed at the transcript and/or exon level have been identified in CD34⁺ cells of *SF3B1* mutant MDS cases, several of which are relevant to the known pathophysiology of patients with ring sideroblasts (22). The global gene expression profiles of MDS patients with *U2AF1* mutations have been shown to be characterized by the downregulation of many genes, including splicing and RNA recognition motif (RRM) genes (23).

Spliceosome mutations are highly specific to MDS and to some extent define distinct clinical phenotypes. For example, *SF3B1* mutations are found in the large majority of MDS patients with ring sideroblasts (subtype RARS and RCMD-RS). *SRSF2* mutations were associated with neutropenia and pronounced thrombocytopenia and *ZRSR2* mutations with isolated neutropenia in the more advanced MDS subtypes in

Table 1 Commonly mutated genes in MDS

Genes	Chromosomal Location	Frequency in MDS	Protein function
Splicing factor genes			
<i>SF3B1</i>	2q33.1	20–28% in MDS >70% in MDS with ring sideroblasts	Spliceosome component important for assembly of the spliceosome complex
<i>SRSF2</i>	17q25.1	12–15%	Spliceosome component important for assembly of the spliceosome complex
<i>U2AF1</i>	21q22.3	7–9%	Spliceosome component important for assembly of the spliceosome complex
<i>ZRSR2</i>	Xp22.1	3–11%	Spliceosome component important for assembly of the spliceosome complex
<i>PRPF8</i>	17p13.3	1–4%	Spliceosome component essential for the catalytic step II in pre-mRNA splicing process
DNA methylation			
<i>TET2</i>	4q24	20%	Dioxygenase that catalyzes the conversion of 5-methylcytosine to 5-hydroxymethylcytosine; required for myelopoiesis
<i>IDH1 IDH2</i>	2q33.3 15q26.1	5%	Dehydrogenase that catalyzes the conversion of isocitrate to α -ketoglutarate; regulates TET2 function
<i>DNMT3A</i>	2p23.3	10%	DNA methyltransferase involved in <i>de novo</i> DNA methylation and coordinated with histone methylation to repress transcription
Histone modification			
<i>ASXL1</i>	20q11.2	11–15%	Histone-binding protein that disrupts chromatin in localized areas enhancing or repressing gene transcription
<i>EZH2</i>	7q36.1	5%	Histone methyltransferase involved in transcriptional repression
Signal transduction and transcription factors			
<i>RUNX1</i>	21q22	10%	Transcription factor critical in hematopoiesis
<i>TP53</i>	17p13	5–10%	Tumor suppressor that regulates apoptosis, cell cycle, DNA repair, senescence, and metabolism
<i>NRAS</i>	1p13.2	5%	GTPase with oncogenic function when mutated and constitutively active
<i>KRAS</i>	12p12.1	2%	GTPase with oncogenic function when mutated and constitutively active
<i>ETV6</i>	12p13.2	2–5%	Transcription factor required for hematopoiesis
<i>EVI1</i>	3q26	1–2%	Transcriptional regulator implicated in hematopoiesis, apoptosis, development, differentiation, and proliferation
<i>JAK2</i>	9p24.1	5% in MDS 50% in RARS-T	Non-receptor tyrosine kinase involved in cell cycle, genomic instability, apoptosis, and mitotic recombination
<i>FLT3</i>	13q12	1–4%	Class III receptor tyrosine kinase; regulates hematopoiesis
Cohesin complex			
<i>STAG2</i>	Xq25	7%	Subunit of the cohesin complex, which regulates the separation of sister chromatids during cell division
<i>RAD21</i>	8q24	1%	Subunit of the cohesin complex, which regulates the separation of sister chromatids during cell division
<i>SMC3</i>	10q25	1%	Subunit of the cohesin complex, which regulates the separation of sister chromatids during cell division
Others			
<i>CBL</i>	11q23.3	2–5%	E3 ubiquitin ligase involved in cell signaling and protein ubiquitination; functions as a negative regulator of signal transduction in hematopoietic cells
<i>SETBP1</i>	18q21.1	4%	Binds the SET nuclear oncogene which is involved in DNA replication and cell division
<i>BCOR</i>	Xp11.4	4%	Interacting corepressor of BCL6, a transcription repressor that may influence apoptosis
<i>CSNK1A1</i>	5q32	7%	Serine/threonine kinase involved in Wnt/ β -catenin pathway

one study (15). *SRSF2* mutations are also strongly associated with chronic myelomonocytic leukemia (CMML) (8). Interestingly, spliceosome mutations, while mutually exclusive with one another, show a strong tendency to co-occur with mutations of specific epigenetic modifiers in MDS, suggesting that abnormalities of these processes may cooperate to give the MDS phenotype (15, 24, 25). *SRSF2* and *ZRSR2* mutations are common in patients carrying mutations of the DNA methylation modifier *TET2*, *SF3B1* mutations co-occur with mutation of the methyltransferase *DNMT3A*, and

U2AF1 mutations have been associated with *ASXL1* or *TET2* mutations (13, 15, 25).

Most recently, somatic mutations and deletions of the spliceosome gene *PRPF8* have been reported in myeloid neoplasms, including MDS and AML (26, 27). The *PRPF8* gene encodes PRP8, the largest and most evolutionarily conserved protein of the spliceosome (28, 29). *PRPF8* mutations were found in approximately 1–4% of patients with MDS, and deletions of this gene were also observed in a similar percentage of patients (26, 27). In common with *SF3B1*,

PRPF8 defects were strongly associated with the ring sideroblast phenotype; however, *PRPF8* defects are associated with a more aggressive MDS phenotype. The association of *SF3B1* and *PRPF8* mutations with the presence of ring sideroblasts suggests that there may be common genes or pathways affected by both *SF3B1* and *PRPF8*. RNA-seq of primary cells from patients with *PRPF8* abnormalities demonstrated consistent aberrant splicing of numerous gene transcripts, including several involved in the hematopoietic pathway and iron metabolism in mitochondria (27).

A recent study has shown that the spliceosome subunit PRPF8 is essential for sister chromatid cohesion in human cells (30), and siRNA-mediated *SF3B1* depletion in HeLa Kyoto cells has been shown to cause an almost complete disruption of sister chromatid cohesion (31). These studies suggest a phenotypic link between splicing and cohesion, and raise the possibility that *SF3B1* mutations could contribute to disease pathogenesis by altering the turnover of cohesin on chromatin, and hence affect chromosome stability, DNA repair, and gene regulation (31). This is intriguing given the recent reports of mutations in components of the cohesin complex in MDS (32, 33). Interestingly, a recent study identified a DNA damage-induced BRCA1 protein complex that contains BCLAF1 and other spliceosomal proteins, including *SF3B1* and *PRP8* (34). In response to DNA damage, this complex regulates pre-mRNA splicing of genes involved in DNA damage signaling and repair and it has been demonstrated that abrogation of members of this complex results in genomic instability. The function of this complex may be impaired in MDS patients with *SF3B1* mutations, with possible downstream effects on the efficiency of DNA damage repair (22, 34).

The prognostic impact of spliceosome mutations has been investigated in various studies, with some conflicting results. *SF3B1* mutations have been shown to be independent predictors of favorable patient outcome in MDS in most, but not all, studies (12, 13, 15, 20, 35). *U2AF1* mutations have been reported to have little impact on survival in some studies (13, 23); however, others have shown that *U2AF1* mutations are associated with poorer survival or high risk of leukemic evolution in MDS (20, 23). *SRSF2* mutations have been shown to be predictive of shorter survival with increased risk of transformation to AML (13, 20, 36).

It is clear that while trends in defining phenotype and prognosis are emerging, detailed functional studies are required to establish the full clinical relevance of spliceosome mutations in MDS disease evolution. One important question that needs to be addressed is how spliceosome mutations lead to a clonal growth advantage in hematopoietic cells.

Epigenetic regulator mutations

CpG island methylation within gene promoters is a major epigenetic transcriptional control mechanism that is frequently

deregulated in human tumors and leukemias (37–39). Various histone modifications that affect gene expression are also commonly observed (38). Changes in DNA methylation patterns in MDS cells may affect not only CpG islands in gene promoters, but also intragenic and intergenic regions, especially enhancers. Epigenetic abnormalities are frequently observed in MDS, including the hypermethylation of genes that control proliferation, adhesion, and other characteristic features of this disorder (38, 40). Several therapies aiming to increase the expression of genes silenced by promoter hypermethylation are currently in use in MDS, including the hypomethylating agents azacitidine and decitabine, important treatment options for high-risk MDS (40, 41).

Several genes involved in the regulation of DNA methylation (*DNMT3A*, *IDH1/IDH2*, and *TET2*) and histone function (*EZH2*, *ASXL1*, and *UTX*) are frequently mutated in MDS (42, 43). Inactivating mutations in the *TET2* gene were identified in MDS and myeloproliferative neoplasms (MPN) from the analysis of recurrent deletions and areas of loss of heterozygosity within a minimal region of chromosome 4q24 (5, 44, 45). Approximately 20% of patients with MDS harbor a *TET2* mutation, and this gene is also frequently mutated in other myeloid malignancies, including CMML and AML (43, 46). *TET2* catalyzes the conversion of 5-methylcytosine to 5-hydroxymethylcytosine in the DNA, possibly leading to loss of DNA methylation (47). *TET2* mutations identified in MDS are thought to result in a loss of catalytic activity. Deletion of *Tet2* in mice leads to increased HSC self-renewal and altered cell differentiation with skewing toward monocytic/granulocytic lineages; *Tet2*-deficient mice develop myeloid malignancies resembling characteristics of MDS and CMML (43, 48). The impact of *TET2* mutations on MDS patient survival is uncertain, as there have been several conflicting studies (43, 46, 49). *TET2* mutations were found to be independently associated with shorter overall survival in patients with MDS after HSC transplantation (50).

The activity of *TET2* is regulated by α -ketoglutarate, which is a metabolite generated by *IDH1* and *IDH2* (3). Mutations of *IDH1* and *IDH2* have been identified in 5–10% of MDS cases and are associated with unfavorable clinical outcome (51–53). *IDH1/2* mutations were found to be mutually exclusive with *TET2* mutations in patients with AML; expression of mutant *IDH* inhibits the hydroxylation of 5-methylcytosine catalyzed by *TET2* (54). Expression of mutant *IDH2* or reduction in *Tet2* expression impaired hematopoietic differentiation in culture and increased stem/progenitor cell marker expression, suggesting a shared pro-leukemogenic effect (54).

DNMT3A is a methyltransferase enzyme that catalyzes the transfer of methyl groups to cytosines in CpG dinucleotides in DNA. *DNMT3A* is required for efficient maintenance of methylation of active chromosome domains (55). *DNMT3A* mutations occur in approximately 10% of patients

with MDS and have been associated with poor survival and more rapid progression to AML (43, 56).

Mutations in the polycomb family member genes *ASXL1* and *EZH2* have been reported in MDS (4, 57). *ASXL1* encodes a chromatin-binding protein involved in epigenetic regulation in hematopoietic cells and plays an important role in recruiting the polycomb repressive complex 2 (PRC2) to specific loci (58, 59). *ASXL1* mutations are common in patients with MDS (occurring in approximately 11–15% of cases) and confer a poor prognosis (9). Mutations of *ASXL1* occur in exon 12 of the gene, and these frameshift and nonsense mutations result in loss of ASXL1 protein expression (58). *ASXL1* mutations are frequently observed in patients with MDS and CMML that transform to AML (60, 61), suggesting a role in disease progression. *ASXL1* mutations are thought to promote myeloid transformation through loss of PRC2-mediated gene repression (58). Hematopoietic-specific deletion of *Asxl1* in mice results in progressive, multilineage cytopenias and dysplasia with increased numbers of hematopoietic stem/progenitor cells, characteristic features of human MDS (62). *ASXL1* knockdown results in impaired myeloid differentiation of human hematopoietic cells in culture (59). *EZH2* is a histone lysine methyltransferase gene that is mutated in 5% of patients with MDS, and it is considered to act as a tumor suppressor in myeloid malignancy (4, 63). *EZH2* mutations are predictors of poor survival in MDS (9). It has recently been shown that combining the Lower-Risk Prognostic Scoring System stratification and *EZH2* mutation status identified a proportion of patients with lower-risk MDS with a worse-than-expected prognosis that may benefit from earlier initiation of disease-modifying treatment (64).

There is evidence to suggest that *DNMT3A*, *ASXL1*, and *TET2* mutations are independently associated with a better response to hypomethylating drugs in patients with MDS (65, 66). In a recent study, *TET2* mutations were shown to predict response to hypomethylating agents (67); response rates were highest in *TET2* mutant MDS patients without clonal *ASXL1* mutations (67). Whether patients carrying mutations in other epigenetic regulators respond differently to the epigenetic therapies employed in MDS remains to be assessed.

Signal transduction and transcription factor mutations

Mutations of genes involved in the RAS signaling pathway occur in approximately 5–10% of patients with MDS, with mutations of *NRAS* being the most frequent (68). *RAS* mutations have been associated with poor prognosis in some, but not all studies of MDS (9, 68–70).

Mutations of genes that encode tyrosine kinases are infrequent events in MDS. However, approximately 50% of patients with the MDS/MPN overlap disease refractory anemia with ring sideroblasts and thrombocytosis (RARS-T)

carry the V617F mutation in the *JAK2* gene, encoding a cytoplasmic tyrosine kinase (71, 72). Internal tandem duplications (ITD) and other activating mutations of the receptor tyrosine kinase *FLT3* are common in AML, but occur less frequently in MDS and have been reported in <5% of patients (68, 73). A mouse model harboring an ITD in the *Flt3* locus was shown to develop a CMML-like disorder (74).

RUNX1, encoding a transcription factor and critical regulator of hematopoiesis, was originally identified as a translocation partner in AML (75). Point mutations in *RUNX1* are frequently found in MDS, occurring in approximately 10% of patients (9, 75). Another transcription factor, *ETV6*, is mutated in approximately 2–5% of patients with MDS (9). Mutations of *RUNX1* and *ETV6* confer a poor prognosis in MDS (9).

Mutations in the tumor suppressor gene *TP53* are frequently found in MDS (5–10% of patients) and are associated with advanced disease, a complex karyotype, and adverse outcome (9, 76, 77). The presence of *TP53* mutations is believed to cause an increase in genomic instability. *TP53* mutations are strongly correlated with abnormalities of chromosome 5 in MDS and confer a poor prognosis (78). *TP53* mutations have also been associated with poor survival in patients with high-risk MDS and AML treated with azacitidine (79).

Cohesin complex mutations

Recurrent mutations in components of the cohesin complex, which is involved in the cohesion of sister chromatids, post-replicative DNA repair, and transcriptional regulation, have been reported in MDS and other myeloid malignancies in the past few years (32, 33). Mutations of *STAG2* were found in approximately 7% of patients with MDS and mutations of *RAD21* or *SMC3* in 1% of cases (32, 33). Mutations of these cohesin genes were mostly mutually exclusive and considered to be loss-of-function. Cohesin defects were associated with poor overall survival, particularly in MDS patients with *STAG2* mutations surviving for more than 12 months (33). In contrast, the presence of cohesin mutations did not influence patient survival in a recent study of AML (80).

SETBP1 mutations

SETBP1, encoding SET-binding protein 1, is a regulator of SET nuclear protein, an oncogene involved in cell division (81). Somatic *SETBP1* mutations were recently identified in several myeloid malignancies, including MDS (82–85). *SETBP1* mutations frequently coexist with $-7/\text{del}(7q)$ and $i(17)(q10)$ (82–84, 86, 87), two cytogenetic markers associated with shortened overall survival in MDS and increased risk of leukemic evolution. Patients with MDS carrying *SETBP1* mutations show a worse prognosis compared to *SETBP1*

wild-type patients (87). The screening of serial MDS/CMML samples shows that *SETBP1* mutations are acquired during leukemic evolution in some patients, suggesting that *SETBP1* mutations may play a role in disease progression (86, 87).

Mutational landscape in MDS

Two recent studies have investigated the complexity of the mutation patterns in large MDS patient cohorts by next-generation sequencing using panels of multiple genes implicated in myeloid malignancy. In the study by Papaemmanuil *et al.* (25), the coding regions of 111 genes were sequenced across 738 patients with MDS. Four genes (*SF3B1*, *TET2*, *SRSF2*, and *ASXL1*) were mutated in >10% of patients, with a further 39 genes carrying driver mutations in <10% of patients. Mutations in genes involved in RNA splicing or DNA methylation were found to occur in early MDS, whereas mutations in genes involved in chromatin modification and cell signaling tended to occur later. Interestingly, somatic, recurrent *TET2* mutations have been found in normal elderly individuals with acquired clonal hematopoiesis (88). Moreover, a recent study has identified *TET2*, *DNMT3A*, *ASXL1*, and *SF3B1* mutations in the normal blood control samples from 2728 patients with cancer, suggesting that these mutations may represent premalignant events that cause clonal hematopoietic expansion (89).

The pairwise associations between genes and the patterns of subclonality identified in the study by Papaemmanuil *et al.* (25) support the concept of genetic predestination, in which early driver mutations (typically affecting splicing factor genes) determine the future trajectory of disease evolution, with distinct clinical phenotypes. Importantly, clonal or subclonal mutations were found to have equivalent prognostic significance. Leukemia-free survival was not significantly different between clonal and subclonal mutations, but deteriorated steadily as the numbers of driver mutation increased (25).

In another study, Haferlach *et al.* (26) screened 944 patients with MDS for mutations in 104 genes. Six genes (*TET2*, *SF3B1*, *ASXL1*, *SRSF2*, *DNMT3A*, and *RUNX1*) were found to be mutated in >10% of cases, with several other genes mutated at a lower frequency, a finding consistent with the results of the study by Papaemmanuil *et al.* (25). A prognostic model based on the status of 14 genes was able to separate patients into four significant risk groups (26).

These studies greatly illuminate the genomic landscape of MDS and support the incorporation of somatic mutation information into MDS prognostic scoring systems.

Impact of gene mutations on clinical features

Several studies have demonstrated that the presence of somatic mutations in specific genes is associated with

distinct effects on cytopenias and blast proportion in MDS. Mutations of *ASXL1*, *RUNX1*, and *EZH2* have been significantly associated with reduced hemoglobin levels, for example (64). Mutation of *RUNX1*, *TP53*, *NRAS*, *ASXL1*, and *U2AF1* has been associated with thrombocytopenia (9, 64), while *SF3B1* mutations have been associated with a normal or increased platelet count (12, 64). Mutations in *ASXL1*, *SRSF2*, *CBL*, *RUNX1*, *WT1*, *IDH2*, *STAG2*, and *NRAS* have all been shown to correlate with an increased percentage of bone marrow blasts (25, 26, 60, 64).

A recent large study, aiming to define genotype/phenotype relationships of clinical relevance, investigated 308 patients with MDS, MDS/MPN, or secondary AML using unsupervised statistical analysis, including the WHO classification criteria and gene mutations (90). It was demonstrated that MDS associated with *SF3B1* mutation (20.8% of cases) is a distinct nosologic entity irrespective of current morphologic classification criteria, while MDS with ring sideroblasts with non-mutated *SF3B1* segregated into different clusters with other MDS subtypes. Mutations of genes involved in DNA methylation, splicing factors other than *SF3B1*, and genes of the RAS pathway and cohesin complex were independently associated with multilineage dysplasia and identified a distinct subset of patients (20.8% of cases). Irrespective of driver somatic mutations, a threshold of 5% bone marrow blasts retained a significant discriminant value for identifying cases with clonal evolution. Co-occurrence of *TET2* and *SRSF2* mutations was strongly predictive of a myeloid malignancy characterized by myelodysplasia and monocytosis, including CMML. These results indicate that a molecular classification of myeloid malignancy is feasible.

Chromosomal abnormalities

Cytogenetic abnormalities are present in approximately 50% of *de novo* MDS and 80% of therapy-related MDS cases. The karyotype is an independent prognostic factor in MDS (91, 92). In a recent study, Schanz *et al.* (92) have analyzed the cytogenetic subsets of >2000 patients with MDS and have proposed a new and comprehensive cytogenetic scoring system that has been subsequently validated within the revised International Prognostic Scoring System (IPSS-R) (93). The IPSS-R has integrated the cytogenetic abnormalities effectively for MDS prognostic risk analysis (93).

Chromosomal monosomies and deletions are commonly observed in MDS. Interstitial deletion within the long arm of chromosome 5 [del(5q)] and monosomy of chromosome 7 or del(7q) [−7/del(7q)] are among the most frequently reported karyotypic abnormalities in this disorder. Other recurrent deletions occurring in MDS include del(20q), del(17p), and del(11q) (94, 95). It remains to be determined whether these chromosomal abnormalities are initiating events leading to the development of MDS. A recent elegant study has shown that in MDS with isolated del(5q),

acquisition of del(5q) preceded diverse recurrent driver mutations in most cases, and this finding is compatible with the del(5q) being the initiating and potentially the only genetic lesion required for the development of MDS with isolated del(5q) (96).

The del(5q), del(7q), and del(20q) are considered to mark the location for one or more genes, the loss of which may affect important processes involved in normal hematopoiesis in MDS and AML. It has been increasingly recognized that gene haploinsufficiency may play a role in the development or progression of cancer, and the importance of this mechanism in the context of myeloid malignancy is supported by studies concerning the 5q- syndrome (97, 98).

The 5q- syndrome and MDS with del(5q)

The del(5q) occurs in approximately 10–20% of patients with *de novo* MDS (94) and in a similar proportion of patients with *de novo* AML. The 5q- syndrome is the most distinct of the MDS and is defined as refractory anemia with the del(5q) as the sole karyotypic abnormality (99). Great progress has been made in the last decade in the elucidation of the molecular basis of the 5q- syndrome (100). The commonly deleted region (CDR) of the 5q- syndrome was identified and narrowed to a ~1.5 Mb interval at 5q32 by Boultonwood *et al.* (101). Reduced gene expression (approximating to haploinsufficiency) of several candidate genes mapping to this interval, including the ribosomal protein gene *RPS14*, *SPARC*, and the serine/threonine kinase *CSNK1A1*, was demonstrated in the CD34⁺ cells of patients with 5q- syndrome (102). A comprehensive mutation screen of all 40 genes in the CDR at 5q32 in a cohort of ten patients with the 5q- syndrome was performed by our group using Sanger sequencing, and no mutations were identified. Recently, mutations of *CSNK1A1* have been reported in a

subset of 5q- cases (approximately 7%) (103). A D140Y mutation of *CSNK1A1* was reported earlier in a single patient with del(5q) MDS (96). Thus, recurrent mutations have now been identified in a gene mapping to the CDR, but only in a small proportion of patients.

In a pivotal study, Ebert *et al.* (98) used a RNA interference (RNAi)-based approach to screen all the genes in the CDR of the 5q- syndrome and demonstrated that haploinsufficiency of *RPS14* is critical in the development of the anemia that characterizes this disorder. Haploinsufficiency of *RPS14* in normal HSC was shown to result in a block in erythroid differentiation, and forced expression of an *RPS14* cDNA in bone marrow cells from patients with 5q- syndrome rescued the phenotype, strongly suggesting that *RPS14* is a 5q- syndrome gene (98).

In another important study, Barlow *et al.* (97) used large-scale chromosomal engineering to generate a mouse model of the 5q- syndrome that displays the key features of the human disease, including a macrocytic anemia. The bone marrow cells of the '5q- mice' show *RPS14* haploinsufficiency and an accumulation of p53 protein and increased apoptosis. Intercrossing the 5q- mice with p53-deficient mice rescued the progenitor cell defect, restoring HSC bone marrow populations (97). This study was the first to suggest that a p53-dependent mechanism underlies the pathophysiology of the 5q- syndrome (97).

It is now well recognized that p53 activation occurs in response to 'ribosomal stress', leading to transcription of its many target genes and resulting in p53-dependent cell cycle arrest or apoptosis (104) (Fig. 1). It has been shown that p53 activation is a common response to decreased levels of ribosomal proteins and that this mechanism underlies the phenotypes of various disorders of defective ribosomal biogenesis, most notable of which are the bone marrow failure syndromes, such as Diamond Blackfan anemia, and now

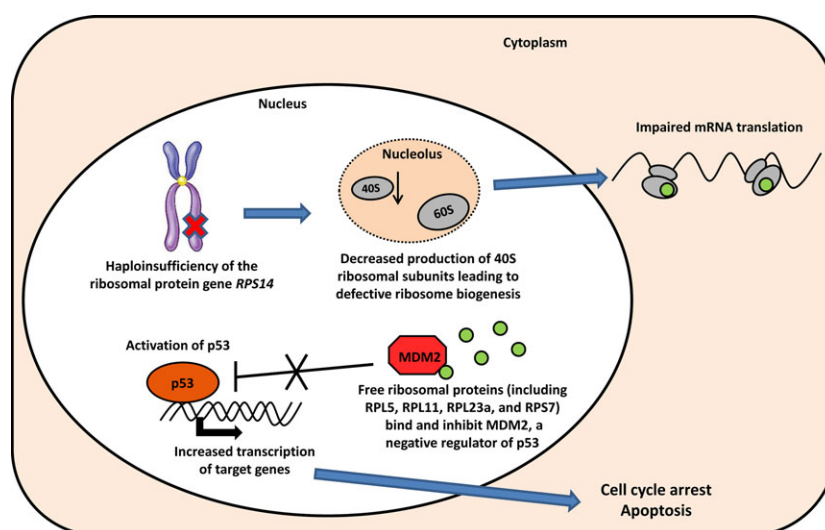


Figure 1 Haploinsufficiency of *RPS14* results in p53 activation in erythroid progenitors.

also the 5q- syndrome (100, 104, 105). Interestingly, there is evidence to suggest that lenalidomide, a drug used to treat patients with the 5q- syndrome (106, 107), acts in part through the promotion of p53 degradation in this disorder (108).

TP53 mutations are common in del(5q) MDS (109, 110). There is evidence suggesting that mutation of *TP53*, resulting in loss of p53 protein function, may be one of the molecular events involved in clonal progression of the 5q- syndrome to AML (110). Using deep-sequencing technology, Jädersten *et al.* (110) have shown that small subclones of hematopoietic cells that carry *TP53* mutations occur at an early disease stage in approximately 20% of MDS patients with del(5q) and that *TP53* mutations are associated with an increased risk of leukemic evolution and may be associated with lenalidomide resistance.

Multiple cooperating genetic events seem to be necessary for the development of the 5q- syndrome (100). The thrombocytosis observed in some patients with 5q- syndrome has been linked to deficiency of two microRNA genes, miR-145 and miR-146a, that map within and adjacent to the CDR, respectively (111, 112). These important studies and the studies on RPS14 shed light on the molecular basis underlying the megakaryocytic and erythroid defects in the 5q- syndrome. A recent study provides evidence indicating that *CSNK1A1*, also mapping to the CDR, plays a central role in the pathogenesis of del(5q) MDS. Heterozygous inactivation of *Csnk1a1* in a mouse model causes HSC expansion and β -catenin activation (103). Thus, haploinsufficiency of *CSNK1A1* may well be the cause of the initial clonal expansion in patients with 5q- syndrome.

MDS patients with del(5q) and additional karyotypic abnormalities and secondary MDS cases with del(5q) have a poor prognosis in contrast to the good prognosis associated with the 5q- syndrome. Lai *et al.* (113) have identified a 1–1.5 Mb CDR at 5q31 in AML and in advanced MDS with del(5q). The AML/MDS CDR is distinct from the CDR of the 5q- syndrome and spans several candidate genes, including the transcription factor *EGR1* (113). Reduced or absent expression of *CTNNA1*, a putative TSG also mapping to the AML/MDS CDR, has been shown in patients with AML or MDS with del(5q) (114). Other genes that map outside the CDRs on 5q, including *NPM1* and *APC*, have been reported as having possible importance in myeloid disorders (100, 115). It is likely that the loss of several genes mapping within the del(5q) has various phenotypic effects and contribute to disease pathogenesis in del(5q) MDS.

Monosomy 7 and del(7q)

The -7/del(7q) is found in approximately 5–10% of *de novo* patients with MDS and in approximately 50% of therapy-related cases. The presence of -7/del(7q) is invariably a bad prognostic marker in MDS and AML (92). Several

CDRs located on 7q have been identified in MDS and AML, including CDRs at 7q22, 7q32–33, and 7q35–36 (116, 117). McNeerney *et al.* (118) have recently shown that *CUX1*, a gene mapping to the CDR at 7q22 and encoding a transcription factor, is expressed at haploinsufficient levels in AML and other myeloid malignancies with -7/del(7q). It is thought that *CUX1* may exert tumor suppressor activity in myeloid cells through the regulation of genes involved in cell cycle control (118, 119). Inactivating mutations of *CUX1* were also found in 3% of patients with MDS/CMML in the study by Papaemmanuil *et al.* (25) [importantly these patients did not show del(7q)], providing evidence that *CUX1* acts as a TSG in myeloid cells. Deletions of chromosome 7q are typically large, and haploinsufficiency of several genes mapping to the deleted region(s) may contribute to disease pathogenesis.

Gene expression profiling in MDS

The development of microarray-based gene expression profiling (GEP) over a decade ago has provided a new powerful tool for investigating the biology and clinical heterogeneity of MDS, and for the identification of new therapeutic targets, predictors of drug response, and prognostic markers in MDS.

An early GEP study on bone marrow CD34⁺ cells of 55 patients with MDS revealed expression profiles similar to interferon- γ -induced gene expression changes in normal CD34⁺ cells (120). CD34⁺ cells from patients with RARS were characterized by deregulation of mitochondrial-related genes, while CD34⁺ cells from patients with the 5q- syndrome showed downregulation of multiple ribosomal genes and genes involved in translation initiation (120). In a subsequent GEP-based study on the CD34⁺ cells of 183 patients with MDS, it was demonstrated that patients with early MDS showed deregulation of immunodeficiency, apoptosis, and chemokine signaling pathways, whereas patients with advanced MDS were characterized by deregulation of DNA damage response and checkpoint pathways (11). Distinct gene expression profiles and deregulated gene pathways were also identified in patients with 5q-, trisomy 8, or -7/del(7q). These deregulated pathways further illuminate the molecular pathogenesis of MDS and may provide new targets for therapeutic intervention (11).

Several parameters that form the basis of MDS classifications and prognostic scoring systems widely used in clinical practice are affected by operator-dependent variation. GEP could enhance current diagnostic and prognostic systems by providing standardized, objective gene signatures. Sridhar *et al.* (121) defined a 'poor risk' signature based on the expression of six genes, which was associated with the risk of transformation from MDS to AML. A classification model has been developed to assess the clinical accuracy of GEP (on total bone marrow mononuclear cells) in distin-

guishing 16 acute and chronic leukemia subclasses, MDS, and non-leukemic conditions (122). While this model proved to be very accurate in subtyping leukemia (93% accuracy), only 50% of MDS samples were correctly classified, reflecting the marked heterogeneity of MDS (122). Importantly, a prognostic classification model has been generated that accurately discriminated patients with a rapid transformation to AML within 18 months from those with more indolent disease, using GEP analysis (10). More recently, Pellagatti *et al.* (123) investigated the relationship between gene expression levels and prognosis using GEP data on CD34⁺ cells from a large group of patients with MDS. A signature, based on expression data from 20 genes, was identified that outperformed other predictors including one which additionally used clinical information. Moreover, the gene signature based on CD34⁺ cells significantly identified a separation of patients with a good or poor prognosis in an independent MDS GEP dataset generated from bone marrow mononuclear cells. This prognostic signature may thus be applicable to bone marrow cells without the need to isolate CD34⁺ cells, enhancing its potential clinical applicability in routine practice (123).

A recent study has investigated the interconnections between mutations, gene expression profiles, clinical variables, and patient outcome in MDS (124). It was shown that the mutational status shapes the gene expression landscape, and deregulated genes associated with the most common gene mutations in MDS were identified. Importantly, a model investigating the influence of mutations and expression changes on diagnostic clinical variables as well as survival showed that the transcriptome was the most powerful predictor of outcome (124). These data suggest that incorporation of gene expression data into MDS prognostic scores may increase accuracy in outcome prediction. For clinical applicability, it would be preferable to develop an assay that can measure the expression levels of prognostic genes using a more accessible platform such as real-time quantitative PCR.

Conclusions and future developments

A wealth of new gene mutations has been identified in MDS, and further work, including functional studies, is required to determine how these gene mutations cooperate to give the MDS phenotype. Other important goals include the determination of the clinical impact of all these mutations on response to therapy and MDS patient survival in large patient cohorts. Sequencing-based studies suggest that multiple mutations may play a role in the progression of MDS to AML, but further work is necessary to fully understand the molecular basis of leukemic transformation in MDS.

It is the opinion of many that in order to move the field forward, new MDS classification and prognostic systems that

include molecular parameters should be achieved. The integration of interconnected streams of genomic, transcriptional, and clinical data into the clinical management of patients with MDS will ensure optimal predictive performance.

The central problem with MDS is their heterogeneity. Technological improvements over time will allow us to achieve a better understanding of the genetic and biological basis for this heterogeneity, and this in turn will lead to improved therapies.

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